

WORK SHEET - 18

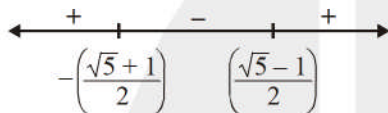
APPLICATION OF DERIVATIVE

Q.1 (B)

$$\begin{aligned} f(x) &= x^3 - 3ax^2 + 3(a^2 - 1)x + 1 \\ f'(x) &= 3x^2 - 6ax + 3(a^2 - 1) \\ &= 3(x - (a + 1))(x - (a - 1)) \\ \text{So } f'(x) = 0 &\Rightarrow x = a + 1 \text{ or } a - 1 \\ \therefore a + 1 &\in (-2, 4) \text{ if } a \in (-3, 3) \\ \text{and } a - 1 &\in (-2, 4) \text{ if } a \in (-1, 5) \\ \therefore a &\in (-1, 3) \end{aligned}$$

Q.2 (A)

$$f(x) = -e^x \left(x + \frac{1 - \sqrt{5}}{2} \right) \left(x + \frac{1 + \sqrt{5}}{2} \right)$$



Thus increasing on $\left[\frac{-1}{2}, \frac{1}{2} \right]$

Q.3 (B)

$$f'(x) = 3x^2 - 6x + 8.$$

$$\text{Also } f(4) = 0 \text{ and } f(1) = 3. \text{ Thus } \frac{f(4) - f(1)}{4 - 1} = -1 = 3c^2 - 12c + 8$$

$$\Rightarrow c = 1 \text{ and } c = 3 \quad \text{Ans.}$$

Q.4 (B)

$$x f(x) = x + C$$

$$f(1) = 1 + C \Rightarrow C = 1$$

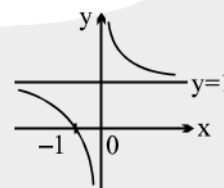
$$f(x) = \frac{x+1}{x}$$

$$f(x) = 1 + \frac{1}{x}; \quad f'(x) = -\frac{1}{x^2}$$

$\Rightarrow f$ is always derivable and decreasing in its domain $\Rightarrow$ monotonic
also f is not bounded.

The graph of $y = f(x)$ is as shown

$y = 1$ and $x = 0$ are the two asymptotes and range is $\mathbb{R} - \{1\}$.



Q.5 (A)

$\therefore f(x)$ satisfies condition of Roll's theorem in $[2, 3]$

and $f(2) = f(3)$

$$\therefore \int_2^3 f'(x) dx = [f(x)]_2^3 = f(3) - f(2) = 0 \quad \text{Ans.}$$

Q.6 (B)

$$f(x) = -e^{-x}(x^2 + 2x + 2) + x$$

$$f'(x) = e^{-x}(x^2 + 2x + 2) - e^{-x}(2x + 2) + 1 = x^2 e^{-x} + 1$$

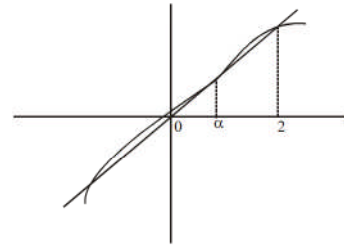
$$f''(x) = x^2 e^{-x} + e^{-x} 2x$$

$$x e^{-x}(2 - x) = 0 \text{ at } x = 0 \text{ or } 2 \text{ POI}$$

now $\alpha \in (0, 2)$

for tangent to cut at 2 distinct points

$$a + b = 2.$$



Q.7 (A)

$$f'(x) = \frac{-1}{(1 + (g(x))^2)} \times g'(x)$$

$$f'(x) > 0$$

Q.8 (B)

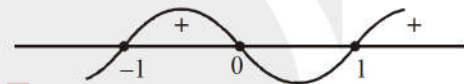
$$f'(x) = 2(x^2 - 1)(2x^2 + 1)^3(2x^2 - 3)^2 \cdot 2x$$

$\therefore$ for maxima

$$f'(x) = \lambda x(x-1)(x+1) \quad \text{where } \lambda = \tan x$$

$\therefore$ f has maxima at $x = 0$

minima at $x = \pm 1$



Q.9 (C)

$$y' = -\sin(x + y) \cdot (1 + y')$$

$$\Rightarrow \frac{-1}{2} = \frac{-1}{2} \sin(x + y) \Rightarrow \sin(x + y) = 1 \Rightarrow \cos(x + y) = 0$$

$$\therefore y = 0 \Rightarrow \sin x = 1$$

$$\therefore x = \frac{\pi}{2}$$

$$\text{So, } P\left(x = \frac{\pi}{2}, y = 0\right) \Rightarrow k = \frac{\pi}{2} \quad \text{Ans.}$$

Q.10 (D)

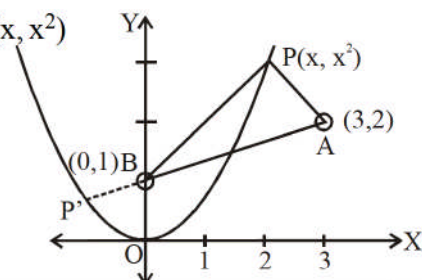
$$f(x) = \sqrt{(x^2 - 2)^2 + (x - 3)^2} - \sqrt{x^2 + (x^2 - 1)^2}$$

Note that 1st radical sign gives the distance between $P(x, x^2)$

and $A(3, 2)$ whereas 2nd radical sign describes

the distance between $P(x, x^2)$ and $B(0, 1)$.

Also (x, x^2) lies on $y = x^2$.



But $PA - PB \leq AB$ for possible positions of P.

Hence $f(x)]_{\max} = \text{distance between AB} = \sqrt{9+1} = \sqrt{10} \Rightarrow \sqrt{M} = 10$. **Ans.**

Q.11 (B)

Q.12 (A)

(i) The equation $P(x) = 0$ has 2 distinct roots and the equation $P'(x) = 0$ has 3 distinct roots.

$$\text{Also, } h(x) = (P'(x))^2 + P(x)P''(x) = \frac{d}{dx}(P(x) \cdot P'(x))$$

So, by Rolle's theorem to $y = P(x) \cdot P'(x)$, we get $h(x) = 0$ has minimum 4 real roots.

(ii) Consider, $P(x) = -k$ has 4 distinct real roots.

$$\Rightarrow -k \in (1, 2) \Rightarrow k \in (-2, -1)$$

$$\therefore [\alpha] = -2, [\beta] = -1, [\gamma] = 0, [\delta] = 2$$

Q.13 (B)

Q.14 (D)

Q.15 (B)

$$(i) \quad 2g'(x) = 2g(x) - 3 \Rightarrow \int \frac{2g'(x)dx}{2g(x) - 3} = \int 1 \cdot dx \Rightarrow g(x) = \frac{3}{2} - \frac{3}{2} \cdot e^{x-1} \quad (\text{As, } g(1) = 0)$$

$\therefore$ Equation of normal at $(1, 0)$ is

$$(y - 0) = \frac{2}{3}(x - 1)$$

$$\therefore y \text{ intercept} = \frac{-2}{3} \quad \text{Ans.}$$

$$(ii) \quad \text{Range of } g(x) = \left(-\infty, \frac{3}{2}\right) \quad \forall x \in \mathbb{R}.$$

$$(iii) \quad \text{Put } x = 1, \text{ in the given functional equation, we get } \int_0^1 g(t)dt = 3 + k$$

$$\Rightarrow k = -3 + \int_0^1 \left(\frac{3}{2} - \frac{3e^{t-1}}{2}\right)dt \Rightarrow k = \left(\frac{3}{2e} - 3\right).$$

Q.16 (BC)

$$\log_2(1 + \sqrt{6x - x^2 - 5}) \geq 0 \Rightarrow 1 + \sqrt{6x - x^2 - 5} \geq 1$$

$$\Rightarrow 6x - x^2 - 5 \geq 0$$

$$\Rightarrow x^2 - 6x + 5 \leq 0$$

$$\Rightarrow (x - 1)(x - 5) \leq 0 \Rightarrow x \in [1, 5]$$

$$f(x) = \int_0^x (3t^2 + 2t + 2)dt \Rightarrow f'(x) = 3x^2 + 2x + 2$$



$\therefore f(x)$ is increasing $\quad \hookrightarrow D < 0$

$$f(x)|_{\min} = L = f(1) = \int_0^1 (3t^2 + 2t + 2) dt = (t^3 + t^2 + 2t)|_0^1 = 4$$

$$f(x)|_{\max} = M = f(5) = \int_0^5 (3t^2 + 2t + 2) dt = (t^3 + t^2 + 2t)|_0^5 = 160.$$

Q.17 (AD)

$$\frac{dy}{dx} = 3ax^2 + 2bx + c \quad \therefore \left(\frac{dy}{dx} \right)_{(0,0)} = c = 1 \quad \dots\dots(i)$$

$$\text{and } \left(\frac{dy}{dx} \right)_{(1,0)} = 3a + 2b + c = 0 \text{ as x-axis is tangent.}$$

$$\therefore 3a + 2b + 1 = 0 \quad \dots\dots(ii)$$

$$(1, 0) \text{ satisfy the curve } \Rightarrow a + b + c = 0 \quad \dots\dots(iii)$$

From (i), (ii) and (iii)

$$a = 1, b = -2, c = 1$$

$$\text{Hence } f(x) = x^3 - 2x^2 + x$$

$$\text{and } f'(x) = 3x^2 - 4x + 1$$

$$f''(x) = 6x - 4$$

Q.18 (AB)

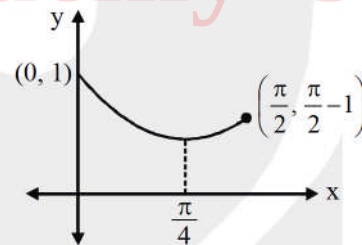
Put $x = \cos \phi$

$$f(\theta) = \int_0^\theta (\sin \theta - \sin \phi) d\phi + \int_\theta^{\frac{\pi}{2}} (\sin \phi - \sin \theta) d\phi$$

$$= 2(\theta \sin \theta + \cos \theta) - \frac{\pi}{2} \sin \theta - 1$$

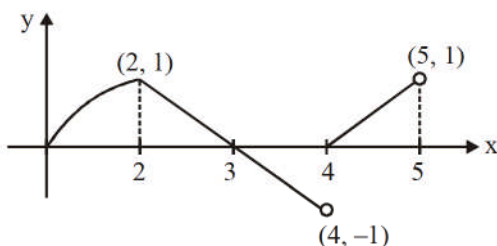
$$f'(\theta) = 2\left(\theta - \frac{\pi}{4}\right) \cdot \cos \theta$$

$$\Rightarrow f(\theta) \text{ decrease in } \left(0, \frac{\pi}{4}\right) \text{ and increase in } \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$



Q.19 (AD)

Answer is obvious from the graph



Q.20 [0004]

$$f(x) = \int_x^{x^2} (t-1) dt$$

$$\Rightarrow f'(x) = 2x^3 - 3x + 1$$

$$\Rightarrow f''(x) = 6x^2 - 3 = 3(2x^2 - 1) > 0 \quad \text{for } x \in (1, 2)$$

$\Rightarrow f'(x)$ is positive in $(1, 2)$

$\Rightarrow f'(x)$ is increasing in $(1, 2)$

$$\therefore f_{\max}(x) = f(2) = \int_2^4 (t-1) dt = 4$$

Q.21 [9]

$f''(x) = 2 \Rightarrow$ Graph of $f(x)$ is concave upward on $(0, \infty)$.

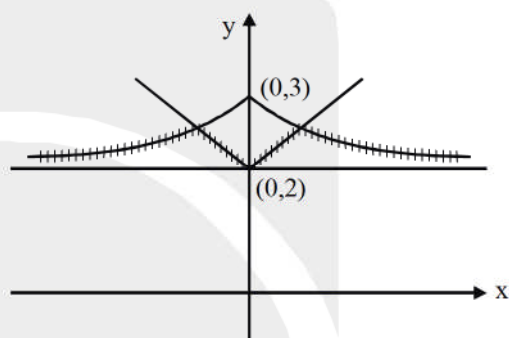
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ and $f'(\infty) \rightarrow \infty$

$$L = \lim_{x \rightarrow \infty} \left(\frac{3x^2}{f(x)} - \frac{3}{(x^2+1)f(x)} - \frac{4f'(x)}{f(x)} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{6x}{f'(x)} - \frac{4f''(x)}{f'(x)} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{6}{f''(x)} \right) = 3$$

$\therefore [L^2] = 9$ Ans.



Q.22 [7]

$$h = p$$

$$k = \frac{4}{3-p} - 1$$

$$PM = CM - PC$$

$$\text{Now, } (CM)^2 = (h-3)^2 + (k+1)^2$$

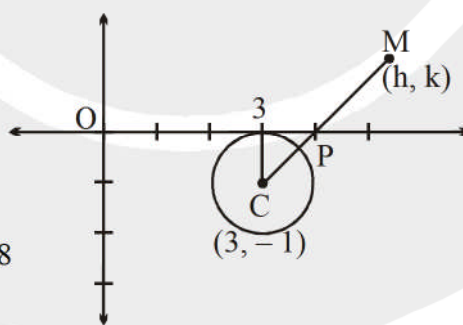
$$(CM)^2 = (p-3)^2 + \frac{1}{(p-3)^2}$$

$$d^2 = (CM)^2 = \left((p-3) - \frac{4}{(p-3)} \right)^2 + 8$$

$$\therefore d_{\min} = 2\sqrt{2}$$

$$\Rightarrow PM_{\min} = 2\sqrt{2} - 1 = \sqrt{8} - \sqrt{1} = \sqrt{a} - \sqrt{b}$$

$$\Rightarrow (a+b) = 7. \text{ Ans.}$$



Aliter: $y = \frac{4}{3-x} - 1$

$$y' = \frac{4}{(3-x)^2}$$

$$\therefore -\frac{(3-h)^2}{4} = \frac{k+1}{h-3} \quad \dots\dots(1)$$

$$\text{and } k+1 = \frac{4}{3-h} \quad \dots\dots(2)$$

From (1) and (2)

$$h = 1, k = 1 \text{ and } h = 5, k = -3$$

$$\therefore d = 2\sqrt{2} - 1 = \sqrt{8} - \sqrt{1} \equiv \sqrt{a} - \sqrt{b}$$

$$\Rightarrow (a+b) = 7. \text{ Ans.}$$

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